LP3 Assignment B5

Roll no: 41168

Problem Statement:

Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset. Determine the number of clusters using the elbow method. Dataset link : Sample Sales Data Dataset (Kaggle) https://www.kaggle.com/datasets/kyanyoga/sample-sales-data

LP3B5: K-Means clustering on sales_data_sample.csv

The number of clusters is chosen with the elbow method.

```
[4]
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

FEATURES = ["QUANTITYORDERED", "PRICEEACH", "SALES", "MSRP"]
MAX_K = 10
print(df.head())
df = pd.read_csv("sales_data_sample.csv", encoding="latin-1")  # file is not UTF-8
df = df.dropna(subset=FEATURES)
X = StandardScaler().fit_transform(df[FEATURES])              # equal weight for every feature
df[FEATURES].describe()
```

Python

```
...    ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

          ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  ... ADDRESSLINE2  \
0   2/24/2003 0:00  Shipped       1         2     2003  ...          NaN
1    5/7/2003 0:00  Shipped       2         5     2003  ...          NaN
2    7/1/2003 0:00  Shipped       3         7     2003  ...          NaN
3   8/25/2003 0:00  Shipped       3         8     2003  ...          NaN
4  10/10/2003 0:00  Shipped       4        10     2003  ...          NaN

            CITY STATE POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME  \
0            NYC    NY      10022     USA       NaN             Yu
1          Reims   NaN      51100  France      EMEA         Henriot
2          Paris   NaN      75508  France      EMEA        Da Cunha
3       Pasadena    CA      90003     USA       NaN           Young
4  San Francisco    CA        NaN     USA       NaN           Brown

  CONTACTFIRSTNAME DEALSIZE CLUSTER
0             Kwai    Small       2
1             Paul    Small       2
2           Daniel   Medium       1
3            Julie   Medium       1
4            Julie   Medium       1

[5 rows x 26 columns]
```

	QUANTITYORDERED	PRICEEACH	SALES	MSRP
count	2823.000000	2823.000000	2823.000000	2823.000000
mean	35.092809	83.658544	3553.889072	100.715551
std	9.741443	20.174277	1841.865106	40.187912
min	6.000000	26.880000	482.130000	33.000000
25%	27.000000	68.860000	2203.430000	68.000000
50%	35.000000	95.700000	3184.800000	99.000000
75%	43.000000	100.000000	4508.000000	124.000000
max	97.000000	100.000000	14082.800000	214.000000

Generate + Code + Markdown

Elbow method

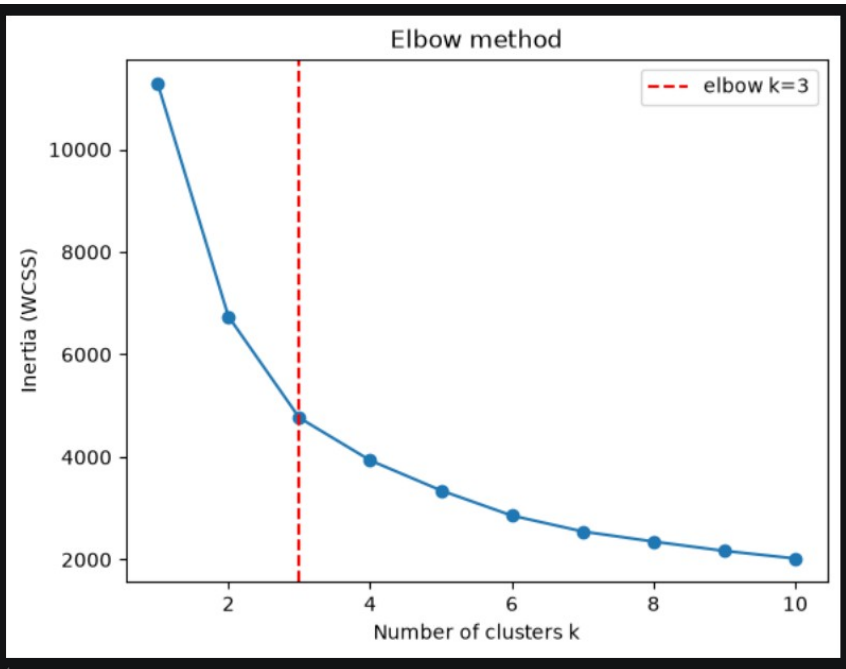
Inertia (within-cluster sum of squares) for k = 1..10. The elbow is the point farthest from the straight line joining the first and last points.

```python
ks = list(range(1, MAX_K + 1))
inertia = [KMeans(k, n_init=10, random_state=0).fit(X).inertia_ for k in ks]

x0, y0, x1, y1 = ks[0], inertia[0], ks[-1], inertia[-1]
dist = [abs((y1 - y0) * k - (x1 - x0) * i + x1 * y0 - y1 * x0) for k, i in zip(ks, inertia)]
best_k = ks[dist.index(max(dist))]

plt.plot(ks, inertia, "o-")
plt.axvline(best_k, ls="--", c="r", label=f"elbow k={best_k}")
plt.title("Elbow method"); plt.xlabel("Number of clusters k"); plt.ylabel("Inertia (WCSS)")
plt.legend(); plt.show()
print(f"Optimal number of clusters: {best_k}")
```

[5] Python




```
Optimal number of clusters: 3
```

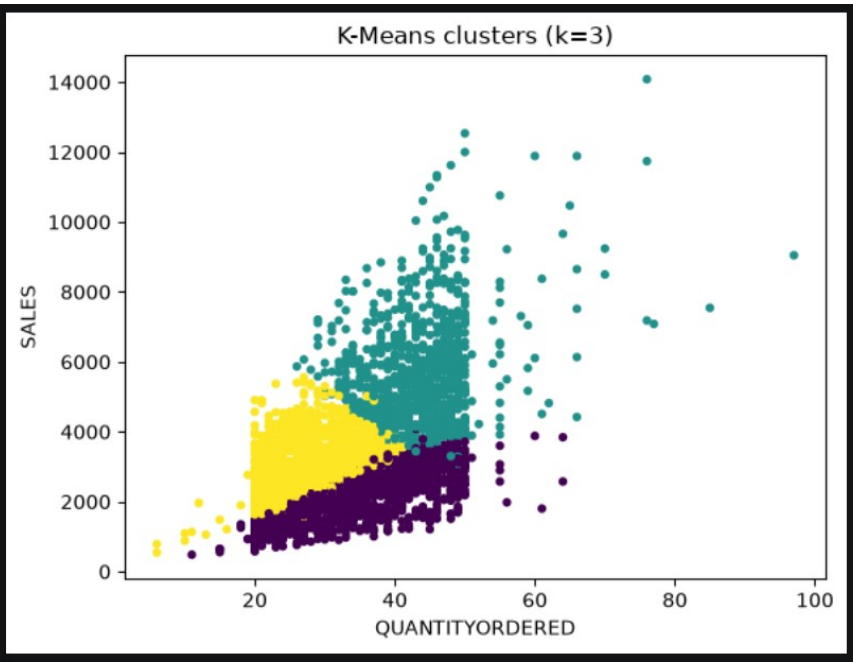
Final K-Means model

```
df["CLUSTER"] = KMeans(best_k, n_init=10, random_state=0).fit_predict(X)

plt.scatter(df["QUANTITYORDERED"], df["SALES"], c=df["CLUSTER"], cmap="viridis", s=12)
plt.title(f"K-Means clusters (k={best_k})"); plt.xlabel("QUANTITYORDERED"); plt.ylabel("SALES")
plt.show()

df.groupby("CLUSTER")[FEATURES].mean().round(2).assign(SIZE=df["CLUSTER"].value_counts())
```

[6] Python



CLUSTER	QUANTITYORDERED	PRICEEACH	SALES	MSRP	SIZE
0	35.21	59.55	2101.08	64.15	975
1	43.72	97.48	5720.08	127.89	852
2	27.60	95.44	3123.06	113.27	996